

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

24th November 2020

Subject CP2B – Actuarial Modelling (Paper B)

Time allowed: 3 Hour 30 Minutes (14.30 – 18.00 Hours)

Total Marks: 100

Exam Requirements

- 1) Read the background document, which describes the scenarios that have been modelled and documented for this project and the work which remains outstanding.
- 2) Read the audit trail which has been written by your colleague, another actuarial student, for the calculations that they performed. This will assist you in following and understanding the calculations performed in the Excel model provided.

You are not required to add to or amend the audit trail.

- 3) Build on the model provided to produce the following additional calculations and comment on the results under each Scenarios comparing them. You should ensure that the additional spreadsheet work which you perform contains appropriate self-checks.

- i) Calculate range of profit and probability of meeting Profit target under Scenario 1 (5)
- ii) Calculate range of profit and probability of meeting Profit target under Scenario 2 (5)

You should assume that your colleague's calculations have been checked and are correct.

- 4) Illustrate in your model the following using suitable charts:

- iii) Profit under normal situation for base and increased capacity compared with profit target under each simulation (3)
 - iv) Profit margin for each hospital per day per patient in normal and outbreak situation with base capacity (2)
- [15]**

- 5) Prepare a summary document of around five to seven pages, capturing the main features and results of the work done by you and your colleague. You can assume that the summary is being prepared for your boss, a senior actuary, who will present the work to the Chief Executive Officer.

Your summary should include the following:

- purpose of the project, data, method and assumptions used by you and your colleague
- results, including charts
- comment on the results, key conclusion and suggested next steps

The summary should cover the full scope of the project, including the modelling already done by your colleague in the spreadsheet provided.

You are not required to add to or amend the audit trail.

Marks available for the summary:

Methodology (including purpose, data, method and assumptions): (25)

Results, including charts (15)

Comment on results and conclusions (20)

Next steps (15)

Drafting (10)

[85]

Background

In the city of Corland, the Health Service company owns five hospitals across the city of Corland. There is a new disease named COVIN-20 which has spread across the world. There are special kits needed for testing of the virus infection in the patients.

The hospital must procure kits in these cities from local area and due to that the cost for each hospital is different for these kits. Each of these hospitals have different capacities and can test only limited number of patients except hospital V. This hospital has a wide facility and is in outer vicinity of the city and hence not preferred by patients in the city.

When the maximum of all 4 hospitals' capacity is reached, people are recommended to hospital V which has huge capacity for the daily patients needing testing.

The cost of testing for each patient and Maximum capacity for each hospital is as below-

Hospital	Cost per patient (\$)	Max number of available kits per day
I	22	50
II	12	500
III	35	200
IV	30	700
V (only if excess patients)	30	NA

The hospitals of Health Service Company are charging \$50 from the patient for testing of the infection. The excess, unused kits, at the end of the day are sent back with no profit and no loss.

The Health Service Company has a profit target of \$20,000 per day from the testing. This profit is mainly used to pay for the staff and other essential items in the hospital.

The CEO of Health Service Company has expressed a concern that there is some uncertainty over the maximum number of kits available with a hospital depending on availability of kits in local area.

The CEO of Health Service Company would like the assessment of whether the company can reach its target profit to allow for the uncertainty of the maximum number of kits in the hospital.

Your boss has suggested using a random number generator to help assess the uncertainty in the maximum number of kits at each hospital. These random numbers are assumed to have a Uniform (0,1) distribution. 100 simulations should be modelled, whereby the adjusted maximum number of kits in a simulation is determined by taking a random number and multiplying it by the maximum number of kits for each hospital on each day.

The CEO has also informed that the patients are **first taken by low cost hospitals and then to higher cost hospitals.**

Your boss has asked that you project the potential range of earnings per day based on 100 simulations for each hospital (I to V) for the various scenarios to be assessed.

Scenario 1

There are 800 expected patients per day for testing.

- (a) The hospitals have same maximum capacity as provided in the table above
- (b) The hospitals I & II were given additional budget to get 20% more kits.

Scenario 2 – Outbreak of disease

There is an outbreak of disease and due to that the expected number of patients per day has gone up to 1,300. For both the above situations (a) and (b), calculate the potential range of earnings per day based on 100 simulations for each hospital (I to V). In this situation Government has also regulated the charges for testing to be \$40 only.

To provide the information for the CEO of Health Service Company, your boss asked your colleague to produce a model to determine the following:

- (i) For Scenario 1(a), determine the number of simulations for which the target profit is reached.
- (ii) For Scenario 1(b), determine the number of simulations for which the target profit is reached assuming the higher capacity of hospital I&II. (Hint: You should multiply the simulations in (i) by 1.2 for this)
- (iii) For Scenario 2, repeat the calculations carried out for (i) and (ii) for the outbreak situation.

Unfortunately, your colleague has fallen ill before being able to complete the modelling required. They have developed a spreadsheet covering step (i) above and produced a first draft of an audit trail for the calculations which have been completed.

Your boss has asked you to complete the outstanding work, using the existing model as your starting point. Your boss would like you to construct, for each of the following, a suitable chart or charts that illustrate:

- Profit under normal situation for base and increased capacity compared with profit target under each simulation
- Profit margin for each hospital per day per patient in normal and outbreak situation with base capacity

Your boss would also like you to prepare a summary document of all the work done (including the work performed by your colleague).

Your boss is out of the office and cannot be contacted for the next three hours. He would like the additional calculations finished and the summary document written up ready for his return, when he will review them and present your findings to the CEO of Health Service Company.

Your colleague has produced an audit trail of his model, which is provided below. An electronic copy of the model is also provided.

You are not expected to include the additional modelling request in the audit trail, but your results should be included in the summary.

Audit Trail

Purpose- The purpose of this project is to be able to calculate the profit of the five hospitals owned by the Health Service company in the city of Corland and check the probability of achieving the Profit Target set by the company under multiple scenarios.

Data- Data has been provided for the no. of kits that a hospital can procure and hence the maximum capacity of a hospital. There is a total expected number of patients in the city in normal and outbreak situation.

The charges per patient have also been provided under normal and outbreak situation-

Income received per patient(\$)	50
Income received per patient at outbreak(\$)	40
Number of patients per day	800
Number of patients per day at outbreak	1,300
Increased capacity multiple for hospitals I&II	1.2

Hospital	I	II	III	IV	V (only if excess patients)
Cost per patient (\$)	22	12	35	30	30
Max number of available kits per day- Base	50	500	200	700	NA
Max number of available kits per day- Higher Capacity	60	600	200	700	NA

Assumptions- The following assumptions have been made in order to calculate the Profit-

- Same amount is charged to each patient
- The total number of patients each day stay constant
- There are no other hospitals in the city where patients can get tested
- Hospital V can always handle as many cases as referred
- Other expenses are ignored
- Taxes are ignored
- Database of hospital is strong and no patients get missed from records

Methodology-

Random numbers between 0 and 1 are generated in Inputs tab itself from row 18 for 4 hospitals using Rand () function.

A new worksheet is created "Capacity- Base"

The simulations generated above are used to calculate the no. of kits available in each hospital per day for 100 simulations from column B to E

This is done by using formula as simulated number * Maximum kits that a hospital can procure. Checks are done that total no. of kits is non-negative and is less than total maximum kits available in all 4 hospitals

New worksheet is added “Scenario 1 (a) to calculate the profit of the company for scenario 1(a). The patients equal to the kits available in hospital are first allocated in hospitals I&II. The sum of these will always be less than total patients as per their max capacity. Then they are allocated to hospital IV depending on the no. of kits in that hospital in column E. This is done by comparing the remaining patients with the no. of kits in that hospital and taking lesser of those. Remaining are then allocated to III in similar way. Any final remaining patients are then allocated to V. This is done because the patients are first taken by low cost hospital.

Total patients are checked to be 800 in column H.

The income for all hospitals combined is calculated in column K by multiplying total patients by amount charged for each patient ($800 \times \$50 = \$40,000$).

Expenses for the 5 hospitals are calculated from column L to P by multiplying cost per patient with no. of patients allocated in that hospital in each simulation. Total cost is calculated in column Q by adding columns L to P.

Profit is calculated in column R by subtracting cost (column Q) from income (column K)

Maximum, minimum and mean profits are calculated in W4,W5 and W6

In Column S, we have checked if the profit is higher than the profit target of 20K.

Number of simulations where the profit exceeds the profit target is calculated in cell W12 and probability of meeting or exceeding profit target is calculated in W13.

http://www.actuariesindia.org/downloads/Year_2020_November_Exam/CP2B_November_2020_Model_for_Exam_Paper.xlsx
